

DATA ACQUISITION TECHNOLOGY

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Executive Summary

Data acquisition (DAQ) technology is the main link between the physical environment and the digital computer system. This paper will study the current situation of data collection technology and compare its evolution from a local analogue recording device to a highly distributed cloud integrated network. By reviewing previous studies and analysing modern data collection architectures, this study focuses on exploring the transformative impact of Industrial Internet of Things (IIoT) and edge computing on high-frequency data collection. This study shows that while modern data collection systems have unparalleled advantages in scalability and real-time processing power, they also bring many problems in data pipeline management, network latency and interoperability. Finally, this paper puts forward strategic suggestions to optimise the infrastructure of the data collection system. The key points include adopting a hybrid edge cloud architecture, optimising the data ingestion process in the data lake and standardising communication protocols.

1. Introduction

Data acquisition (DAQ) refers to the process of measuring electrical or physical phenomena such as voltage, current, temperature or pressure, and then converting the measurement results into digital format for calculation, processing and storage [11]. The standard data acquisition system consists of three main components including sensors and converters, signal adjustment circuits and analogue-to-digital converters (ADC). The sensor will be responsible for converting physical parameters into electrical signals. Then, the signal processing circuit will further optimise these signals to achieve accurate measurement. Finally, the analogue-to-digital converter will convert continuous electrical signals into discrete digital values for calculation processing. The sensitivity, accuracy and calibration of these sensors affect the overall accuracy of the data acquisition system. At the same time, the performance of the ADC depends on its resolution, that is, the number of bits used to represent the value, the minimum change that can be detected in the signal, and the sampling rate [11].

Today's DAQ, especially in this era of artificial intelligence, is no longer just a passive manual recording tool. It has become the main data input point in the complex data engineering process. It supports automation, machine learning and predictive maintenance. In modern industrial systems, DAQ is a key step in connecting the physical machinery and operational technology (OT) of factory workshops with data management and information technology (IT) for enterprise applications [11]. This paper will review the technological improvements of the data acquisition system, assess the current challenges faced by high-frequency data acquisition, and propose feasible suggestions for the modernisation of the data acquisition system.

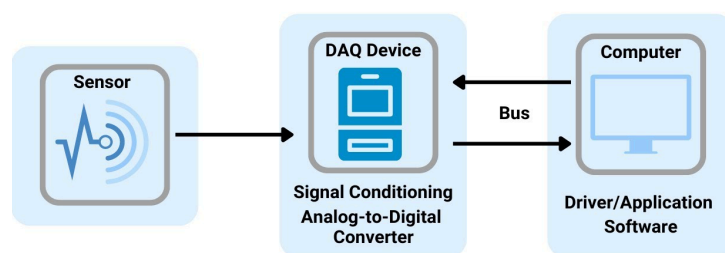


Figure 1.1 Data Acquisition Process

2. Literature Review

The historical development of DAQ system has been well recorded in industrial engineering as well as computer science literature. Early systems mainly relied on the centralised wired architecture using standard buses such as PCI, VXI. Although these architectures are very reliable in localisation, they lack the scalability required for modern distributed systems [6].

According to recent research, the paradigm shift to wireless sensor network (WSN) and industrial Internet of Things (IIoT). Recent research pointed out that the industrial Internet of Things has fundamentally changed DAQ, and is pushing IP-based connections to the sensor level and establishing a large-scale distributed data generation node network [3]. A prominent theme of contemporary research is the integration of edge computing and DAQ pipeline. Recent research also recorded how moving initial data processing such as rapid Fourier transform (FFT) for vibration analysis, noise filtering and initial anomaly detection directly to edge devices greatly reduces the bandwidth required to transmit data to a central cloud server [10]. Besides, the research on the DAQ in intelligent manufacturing often refers to the necessity of real-time response. The traditional pure cloud processing causes unacceptable delays to key control loops which makes edge-level DAQ processing indispensable [12].

3. Analysis and Discussion

3.1 Architectural Transition and Edge Analytics

The most important progress in DAQ at present is to integrate edge analysis into the data pipeline. In the traditional setup, the original high-frequency data was continuously transmitted to the central server. The modern smart DAQ nodes process these signals locally. By running the lightweight ML model directly on the DAQ hardware, the system can detect abnormalities in real time. Only the actionable insights, trigger event logs or aggregates communicate over the network which saves bandwidth and allows real-time automatic replies [7].

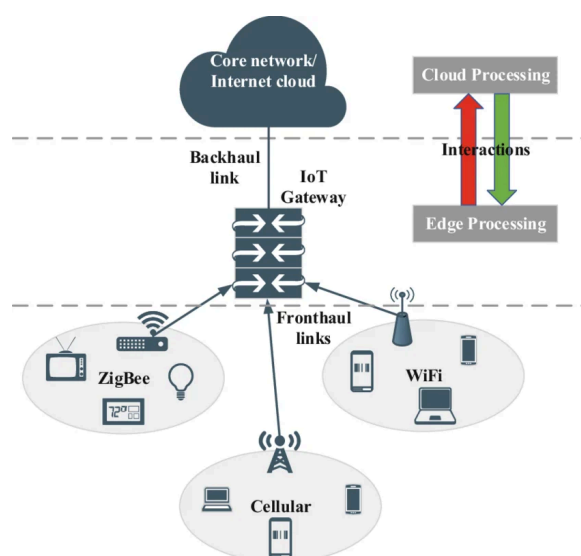


Figure 3.1 Edge-cloud Collaborative System Architecture in IoT Systems

3.2 Interoperability and Protocol Standardisation

As organisations are using sensors from many suppliers under complex circumstances, data isolation has become a major bottleneck. Switching from proprietary supplier agreements to open standards is important to establishing a unified data engineering pipeline. The Message Queue Telemetry Transmission (MQTT) protocol is very effective for low-bandwidth Internet of Things telemetry but the Open Platform Communication Unified Architecture (OPC UA) provides a powerful and secure framework for industrial automation and complex DAQ networks [4].

3.3 High Velocity Data Management

High-speed data acquisition systems, especially those used for automobile testing or power grid monitoring, can generate several megabytes of data per day. Traditional relational databases (RDBMS) cannot meet the requirements of such processing speed. Modern data collection systems require optimised data input pipelines, which can be connected to time series databases (TSDB) or specially designed distributed data lakes to process large loads and sensor data with time stamps.

3.4 Synchronisation in Distributed Systems

When aggregating data from hundreds of scattered DAQ nodes, absolute time synchronisation must be achieved in order to reliably associate each event. Time-sensitive networks (TSN) and precise time protocols (IEEE 1588) can achieve microsecond-level synchronisation in Ethernet-based data acquisition networks, which is important for training precise prediction models [4].

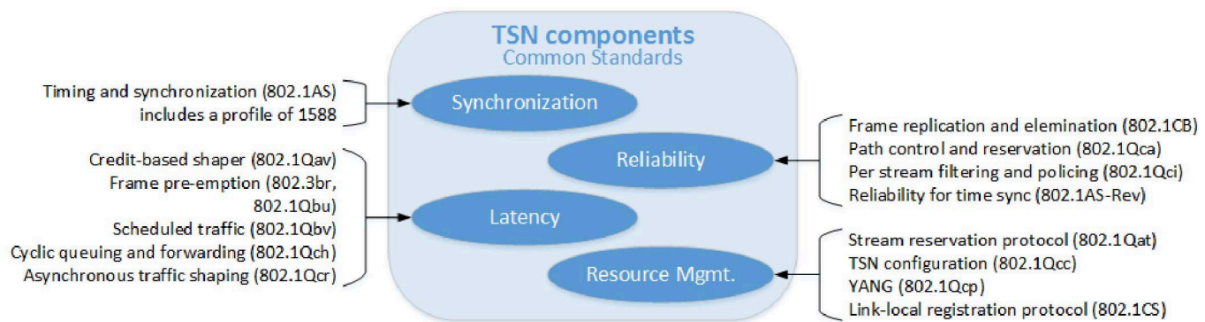


Figure 3.2 TSN Components Common Standards

3.5 Comparative Performance: Batch vs. Stream Ingestion

In modern data engineering, selecting a suitable ingestion method is a strategic decision to balance the "data freshness" against the operational overhead. The batch processing remains the standard for high-volume historical analysis, while stream ingestion become important for systems to require immediate response to live events [9].

Feature	Batch Ingestion	Stream Ingestion
Primary Goal	Build for stability	Build for Responsiveness
Data Bound	Bounded (Discrete windows)	Unbounded (Continuous flow)

Feature	Batch Ingestion	Stream Ingestion
Data Latency	High (Minutes to Hours/Days)	Very Low (Milliseconds to Seconds)
Typical Tools	Apache Hadoop	Apache Kafka, Spark Streaming, Flink
Ideal Use Cases	Financial reporting, ML training, historical trends	Fraud detection, system monitoring, personalization

Figure 3.3 Comparison Table of Batch and Streaming Ingestion

3.6 Challenges in Data Quality and Observability

The transition to high-throughput data acquisition introduces friction between operational speed and data integrity [5], [9]. In a modern environment like Industry 4.0, managing continuous data streams from different sources will bring four different technical problems:

First is the Inherent Sensor and IoT Data Noise problem. Data collected from the physical layer (sensors, RFID) is usually characterized as noisy, incomplete or biased [5]. Errors such as missing values or abnormal values in the process of data acquisition may be caused, leading to scheduling errors, and if the data is not cleaned up immediately, the invalidation of "machine learning applicability" will also be caused.

The second problem is Silent Drift and Backpressure. In streaming engines like Apache Kafka, failures are often non-binary. "Silent correctness drift" is caused when incompatible mode changes or repeated events destroy the data lake, while the dashboard is still displayed as active [8], [9]. In addition, lag accumulation will be caused by "backpressure". That is about when the rate of receiving events exceeds the processing capacity, which makes time-sensitive data irrelevant in operation.

The third problem is The Observability Shift. Traditional acquisition relies on binary "job health" (success/failure), but continuous monitoring of "system health" is required by real-time collection. Nowadays, complex indicators like consumer delay, throughput stability and delivery guarantee must be tracked through effective observability to ensure credibility throughout the enterprise pipeline[9].

The last problem is the problem of Inconsistency Across Functional Silos. When multiple departments obtain data at different speeds and times, information silos will occur. This will lead to differences, that is, different parts of the ERP system will show different inventory levels or different financial reports for the same physical event, making the creation of a "Single Version of the Truth" complicated [2], [5].

4. Recommendations

4.1 Implementation of a Layered Hybrid Architecture

In order to solve the inherent contradiction between data acquisition speed and data authenticity, organizations should adopt a hierarchical hybrid architecture and divide it according to the specific time sensitivity of workloads. This method uses the real-time operation layer to handle instant operation trigger events like anomaly detection and real-time monitoring. While the separate truth layer is used for regular batch processing of authoritative

and authenticated data sets [9]. By implementing shared pattern contracts and automatic deviation detection between these layers, engineers can ensure that real-time operational insights are fully synchronized with long-term and compliance reports [9].

4.2 Deployment of High-Throughput Processing Engines

Technical stacks must be carefully selected to manage continuous, high capacity data flows common in Industry 4.0 environments. Since Apache Kafka has a persistent, log-based architecture, it is recommended as the main intake pillar because it can ensure that strict data sequence retention can be ensured even when the message volume is expanded. In the processing stage, Apache Spark Streaming has a significant advantage over the traditional batch system. It adopts a memory architecture and has an average delay of as low as 1.390 seconds [8]. In addition, these modular microservices are coordinated through Kubernetes, which can independently expand each data ingestion component according to the fluctuation of sensor density and ensure high availability and fault isolation [8].

4.3 Integration of DQaaS and Perpetual Observability

Data integrity is maintained by transferring quality management to data collection points and by using Data Quality as a Service (DQaaS) and comprehensive observability [2]. By automatically cleaning and verifying data when it arrives, the "silent correctness deviation" can be prevented by DQaaS, as noise values in the sensor flow are removed before reaching the analysis layer. Moreover, the transformation from binary "job success" checks to continuous monitoring of "system health" indicators like consumer delay, throughput stability and delivery assurance must be realized through observability [9]. Finally, the implementation of Role-Based Access Control (RBAC) ensures that these continuous data flows are kept secure during the transmission phase so that global regulatory requirements are met [1], [2].

5. Conclusion

Data acquisition technology has developed from simple localised measurement equipment to highly complex distributed networks, as the basic data layer of modern industrial and scientific processes. The combination of improved ADC, edge computing and standardised wireless protocols has achieved unprecedented capabilities in real-time monitoring, machine learning integration and predictive maintenance. However, the success of modern DAQ deployment depends on the ability of the organisation to design powerful data pipelines, which can process a large amount of data, achieve microsecond synchronisation, and strictly prevent network threats. Industries that implement hybrid computing architectures and are committed to open interoperability standards can make full use of DAQ technology to promote efficiency, automation and data-driven decision-making.

References

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Appendix / Logbook

Date	Activity	Tool	AI Prompt	Verification Steps	PIC
25/3/26	Doing research on DAQ concepts and analysed some journals	Google Scholar, NotebookLM	"What is DAQ", "What are sensors and transducers", "What is the function of DAQ in Computer Science", "DAQ uses in AI"	Verified the answers with Google, analysed with academic papers with Google NotebookLM	Lau Yan Kai
29/3/26	Investigated core DAQ concepts and collected academic sources.	Gemini, Google	"Explain Data Acquisition technology simply but in detail," "Search for real academic references for DAQ in Industry 4.0"	Cross-referenced AI-generated definitions with peer-reviewed articles from MDPI and ResearchGate.	Lee Yin Shen
30/3/26	Generated drafts for Intro, Analysis, and Recommendations.	Gemini, NotebookLM	"Generate a draft for the Introduction using APA style," "Create a comparison table for batch vs stream processing based on the resources"	Validated technical metrics (e.g., 1.390s latency for Spark) against 2025/2026 industry benchmarks.	Lee Yin Shen, Lau Yan Kai
31/3/26	Conducted literature search and compiled academic references	Google Scholar, Scopus, IEEE Xplore	Search for academic papers using keywords like "Industrial IoT DAQ", "Edge Computing sensors", and "OPC UA standardisation"	Cross-checked the downloaded papers' DOI link and verify with academic websites of the journals	Lau Yan Kai, Lee Yin Shen
2/3/26	Expanded the introduction and literature review section with more advanced technical concepts	Google Scholar, NotebookLM	Synthesised literature to draft the Analysis section, specifically focusing on how DAQ acts as the critical data ingestion point for scalable data lakes	Verified the newly added technical concepts and checked with the thesis	Lau Yan Kai
3/3/26	Summarise the literatures and drafted conclusion for the report, and proofreading	NotebookLM	Summarised the core findings from the analysis and literature review to draft the conclusion	Reviewed conclusion to make sure answered the paper main objectives and aligned with the executive summary	Lau Yan Kai

4/4/26	Reviewed and adjusted the report layout and ensured all sections are integrated.	Gemini, NotebookLM	"Adjust the overall format of the report," "Review the rubric to ensure all sections meet all requirements."	Final read-through against the Assignment Instructions to ensure all sections meet the requirements.	Lee Yin Shen
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